

AI-Enabled Detection of Exoplanets

Theory, Mathematics, Algorithms and Practical Challenges

1. Project Overview

Pipeline: FITS Loading → Preprocessing → Box Least Squares (BLS) → Parameter Estimation → AI Classification → Visualization → Reporting.

2. Mathematics

Normalized flux: $F_{\text{norm}} = F / \text{median}(F)$. Transit depth $\delta = \text{median}(\text{flux}) - \text{min}(\text{flux})$. Approximate relation: $\delta \approx (R_p/R_*)^2$.

3. Preprocessing

Remove invalid QUALITY points, remove NaNs, normalize flux. Complexity $O(N)$.

4. BLS

Searches thousands of trial periods and transit durations for the box-shaped brightness drop characteristic of planetary transits. Phase folding aligns repeated events. Highest BLS power identifies the best candidate.

5. Phase Folding

$\text{phase} = (\text{time mod period}) / \text{period}$. Correct periods align all transits into one dip.

6. Parameter Estimation

Estimate period, duration, depth, SNR, planet radius and semi-major axis using simplified transit physics and Kepler's Third Law.

7. AI Module

Random Forest classifier trained on NASA Kepler features (period, depth, duration) predicts Planet Candidate vs False Positive.

8. Complexity

Loading $O(N)$; preprocessing $O(N)$; BLS roughly $O(N \times M \times D)$ for N samples, M trial periods and D durations; sorting K candidates $O(K \log K)$.

9. Real-world Challenges

Noise, stellar variability, eclipsing binaries, cosmic rays, missing observations, instrumental systematics and extremely shallow transits.

10. Why BLS?

Transits are box-shaped rather than sinusoidal, making BLS the preferred algorithm over FFT/Lomb-Scargle for this application.

11. Future Work

Support more FITS formats, detrending, CNN-based classification, uncertainty estimation, richer GUI and automatic report generation.